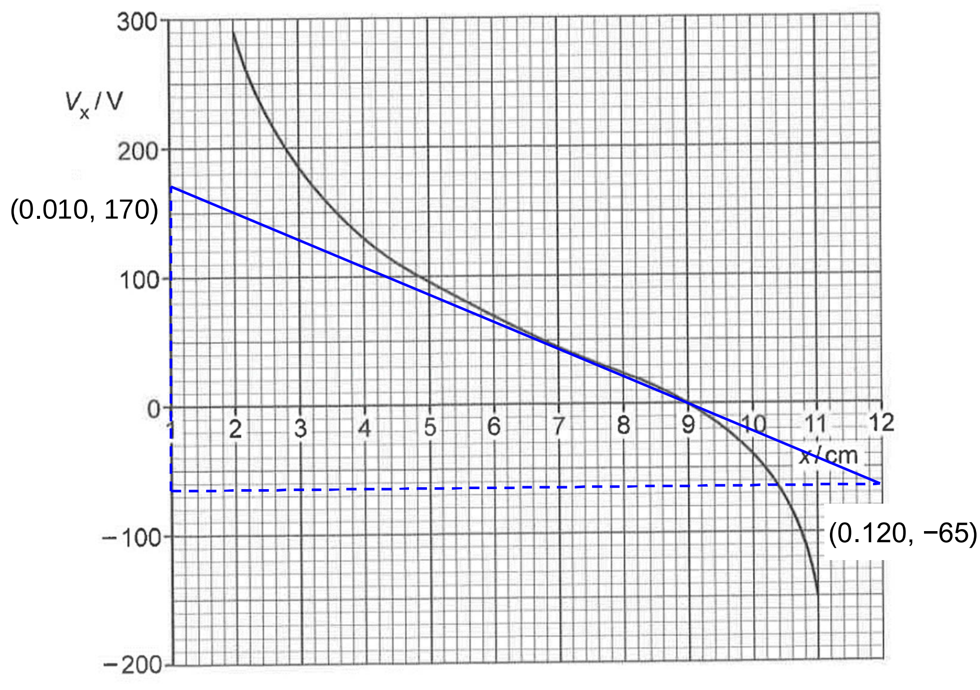
	$= 0.50\ \Omega$
7(a)	similarity: Gravitational potential and electric potential are both inversely proportional to distance from the point mass and point charge respectively. difference: The difference between gravitational potential and electric potential is that the gravitational potential produced by a point mass is always negative, whereas the electric potential produced by a point charge can be either positive or negative.
7(b)(i)	At x = 9.0 cm, the electric potential is zero, which implies the sum of the electric potentials due to point charges A and B is zero. Hence, the electric potential due to one charge is positive, while the electric potential due to the other charge is negative. Hence, the charges have opposite signs. *Note: Both sections of the graph or the zero-point has to be referred to in the explanation.*
7(b)(ii)	At x = 9.0 cm, resultant electric potential of charges A and B = 0 V, implying the potentials are of the same magnitude. i.e. $V_A + V_B = 0$

No.	Solution
	$\dfrac{Q_A}{4\pi\varepsilon_0 r_A} - \dfrac{Q_B}{4\pi\varepsilon_0 r_B} = 0$ $\dfrac{Q_A}{r_A} = \dfrac{Q_B}{r_B}$ $\dfrac{Q_A}{Q_B} = \dfrac{r_A}{r_B}$ $= \dfrac{9.0}{3.0}$ $= 3.0$ *Note: The question asks for explanation of working. Attempting to determine the two charges by using two points on the line and simultaneous equations, while valid, is an extremely difficult approach and is seldom correctly executed.*
7(b)(iii)	 **Fig. 7.2** From Fig. 7.2, gradient coordinates (0.010, 170) and (0.120, -65) of tangent at x = 7.0 cm. Electric field strength,

No.	Solution
	$E = -\frac{\mathrm{d}V}{\mathrm{d}x} = -\frac{-65-170}{0.120-0.010}$ $= -\frac{-235}{0.110}$ $= 2.136 \times 10^{3}$ $\approx 2.14 \times 10^{3}$ N C^{-1}
8(a)	Magnetic flux density is defined as the force per unit length per unit current acting